

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Software Engineer with a strong background in Computer Science and a specialization in Backend Development and Distributed Architectures. Proficient in Golang and container ecosystems such as Docker, with proven full-stack capabilities in React and TypeScript. Passionate about deep engineering topics, including compilers and network protocols, as well as building scalable products. Not afraid to learn new technologies.

Core Competencies

- **Backend & Languages**: Golang, TypeScript/JavaScript, Java, Python, Kotlin, SQL
- **Distributed Systems**: Docker, NATS, Apache Kafka, Apache APISIX, event-driven microservices
- **Full-Stack Development**: React, PostgreSQL, REST APIs, OAuth 2.0/JWT
- **LLM/AI Engineering**: LangChain, LangGraph, Gemini API, OpenAI API, RAG pipelines
- **Tools & Practices**: Git, GitHub Actions, Docker Compose, Testcontainers, TDD

Professional Experience

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
R&D Developer — *Feb 2026–Jun 2026*

- Implemented LLM-based improvements to a search engine
- Built a LangGraph agent converting natural-language questions into validated, read-only SQL to resolve data-access policies in an industrial data-governance/sovereignty platform
- Applied safety-conscious agentic design: schema selection, structured SQL generation, execution validation, retry-on-error

Instituto Tecnológico de Informática (ITI) — **Valencia, España**
Full-Stack Developer Intern — *Feb 2024–Jun 2024*

- Developed and maintained the corporate personnel management application (Intranet)
- Implemented new frontend features using React and TypeScript, improving internal usability
- Contributed to backend development, ensuring smooth integration between services

Education

Polytechnic University of Valencia (UPV) — **Valencia, España**
Master's Degree in Computer Engineering (MUIINF) — *2024–2026*
Completed (Sep 2026), final grade 8.8/10. Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.

Polytechnic University of Valencia (UPV) — **Valencia, España**
Grado (BSc) in Ingeniería Informática, Software Engineering track — *2020–2024*
GPA: 8.88/10. Thesis: "ValenMove - Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (certified B2, confident conversational).

References

- Available upon request.